



Adaptive robust control of multi-DOF hydraulic manipulator with precise online load estimation

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ABSTRACT

Hydraulic manipulator shows vast application potential in heavy load working conditions. However, achieving high control precision in these devices is notably more challenging compared to electric manipulators, due to the complexities introduced by uncertainties and high-order nonlinear dynamics. The presence of heavy unknown payload further reduces control accuracy. In this paper, a new load estimation method based on direct/indirect adaptive robust controller (DIARC) is proposed to facilitate online payload estimation and compensate the impact of the unknown payload. To compensate for the high-order dynamics, backstepping strategy is utilized. A modified recursive least squares adaptive law is also developed to realize, precise, real-time payload estimation under dynamic conditions. By incorporating this accurate load mass estimation into the control strategy, an improvement in overall control performance can be achieved. The effectiveness of this enhanced controller with load mass estimation is initially verified through simulations conducted in MATLAB. The close-loop control performance is further analyzed and validated on a four-degree-of-freedom hydraulic manipulator. The experimental results indicate that, under dynamic scenarios, our proposed control method succeeds in achieving precise online load mass estimation, achieving an average estimation error of 2.8% for a 7.5 kg payload. Furthermore, enhanced control accuracy is achieved compared to traditional controllers, with the maximum tracking error being only 0.69° during the simulated operation scenario. This research provides a viable solution for precision control and load estimation in hydraulic manipulators, eliminating the need for costly force/torque sensors.

1. Introduction

Hydraulic driven systems are extensively utilized in various industrial sectors, including oil and gas production, mining industries, civil engineering industries, and aerospace [1–3]. As industrial automation continues to advance, the demand for precision in various work settings, such as collaborative pipeline assembly and equipment maintenance, is on the rise [4]. Multi degree-of-freedom (DOF) hydraulic manipulators, serving as intelligent hydraulic equipment, are increasingly crucial in these varied environments [5]. However, their application remains constrained due to the lower control precision they offer compared to electric manipulators [6,7].

Compared with other hydraulic systems, hydraulic manipulators necessitate enhanced control accuracy to effectively handle more complex

tasks. However, several key factors severely impair its control effectiveness, with one significant factor being its intricate dynamic characteristics. The hydraulic system, comprising components like proportional valves and servo valves, often introduces high-order nonlinearities, including Coulomb friction, viscous friction, and various unknown external disturbances [8–10]. These complexities pose a significant challenge for PID controllers in achieving high control accuracy [11,12]. To improve the control accuracy of hydraulic system, many researches were carried out. Shen et al. [13] proposed an adaptive integral terminal sliding mode controller to guarantee the robustness of double-variable hydraulic transformer and effectively improved the control effect. Chen et al. [14–17] proposed an adaptive robust motion control theory to deal with the control of single-rod hydraulic actuators and the parameter uncertainty and uncertain nonlinearity of

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cylinder control systems. Yao et al. [18–21] conducted extensive research on the parameter uncertainty of hydraulic systems and combined the RISE-based controller with online parameter adaptive law to effectively suppress parameter uncertainty and reduce tracking errors of hydraulic systems. Hou et al. [22] proposed a force-free control method for compensating for friction, gravity, and inertia of hydraulic Stewart manipulators. By using separate compensation and complete compensation methods, the force-free control of the Stewart mechanism was achieved and good control effects were obtained on parallel robots. While the aforementioned methods have achieved good control performance on the single hydraulic actuator platform, more complex control challenge remains on the multi-DOF hydraulic manipulators which is also limited by joint coupling, system inertia, fluid dynamics and other factors [5,23]. Mattila et al. [24–26] proposed the virtual decomposition control (VDC) method for multi-DOF hydraulic manipulators. Ahn et al. [27] proposed a sliding mode control algorithm using a finite-time disturbance observer for a 3-DOF hydraulic manipulator considering the actuator dynamics and the joint tracking errors were guaranteed to converge to zero in finite time under the strong assumption that the uncertain joint dynamics were time invariant. The aforementioned methods have achieved excellent control performance, but few studies focus on the influence of payload on control accuracy.

While the impact of unknown load mass attached to the end effector is often overlooked, it is another critical factor affecting control precision. In many demanding and harsh operational environments, equipping end effectors with force/torque sensors for precise load mass determination proves difficult. Considering that the load on the hydraulic manipulator is not only time-varying but also typically substantial, this greatly impacts the dynamic behavior of the manipulator. Consequently, there is a pressing need for precise online load estimation that can effectively compensate for this unknown load. Furthermore, such online load estimation is essential in sectors like workpiece handling and cargo sorting, where the awareness of payload mass is crucial. Many studies investigated the unknown load changes of machinery. Lin et al. [28] developed a load sensing system based on displacement adaptive and variable speed control for electric hydraulic excavator, considerably improving its control performance. Duan et al. [29] proposed a variable structure robust control method based on equivalent control and a feedforward compensator for robots with load uncertainty, achieving better control effects under the condition of no force sensor at the end effector. Sun et al. [30] analyzed a novel natural adaption law which achieved accurate control for redundant manipulators when picking tools with unknown length, direction, and mass, and the parameter estimation speed was also increased. Zeng et al. [31] and Colomé et al. [32] explored methods for estimating the load conditions of electric rigid manipulators. Bartlett et al. [33] and Tessari et al. [34] considered the parameter uncertainty of load mass for single-rod hydraulic knee prostheses. Renner et al. [35] proposed online payload estimation method and achieved high estimation performance in earthmoving excavator. However, more research on load compensation and estimation of hydraulic manipulator is needed.

In this paper, we introduce a Direct/Indirect Adaptive Robust Controller (DIARC) to achieve real-time load estimation and improve the control precision of hydraulic manipulators. This controller is specifically designed to mitigate the effects of model nonlinearity and unknown external disturbances, while simultaneously maintaining system stability. Additionally, the load is integrated into the model as a state variable, and a parameter estimation method is employed to ensure accurate load estimation. Through this approach, the controller and load estimator are combined, allowing for a complementary enhancement in both control and estimation accuracy. As well, this strategy not only achieves computational efficiency but also eliminates the necessity for installing force or torque sensors. It has considerable value for mining machinery, sorting machinery, hanging machinery and other industries that calls for precision control and load recognition.

The paper is structured as follows: First, the dynamic modeling of the manipulator is expressed in Section 2. Next, the control law and parameter estimation algorithm are described in Section 3. Section 4 shows the results obtained from simulation. Then in Section 5, the

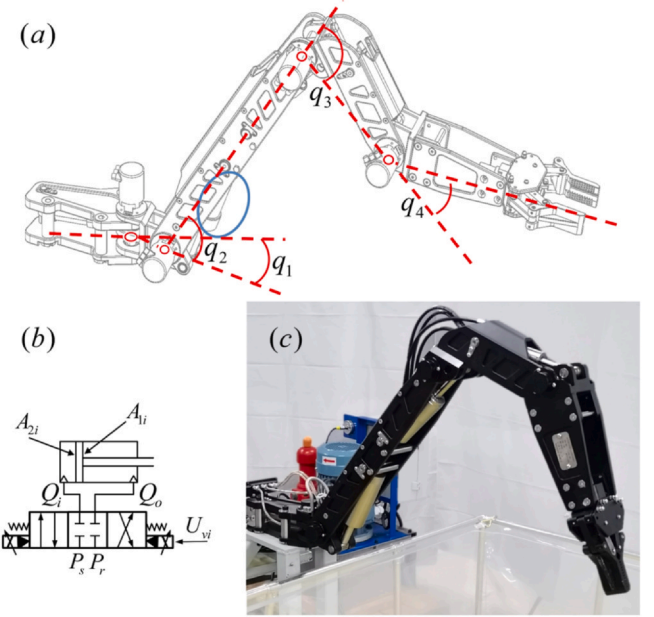


Fig. 1. Schematic of the hydraulic manipulator: (a) Simplified diagram, with the hydraulic cylinder highlighted within the blue frame; its detailed structure is shown in (b), (c) Photograph of the manipulator used in the experiments.

experiment setup is introduced. Following this, in Section 6, the experiment results are discussed before drawing a conclusion in Section 7.

2. Problem formulation

2.1. Dynamic modeling of hydraulic manipulator

The schematic diagram of a multi-DOF manipulator is shown in Fig. 1. For the convenience of dynamic modeling, it is supposed that the centroid of each link is at its center. Therefore, the dynamic model of a n -DOF manipulator is established using Lagrange method as:

$$\mathbf{M}(q)\ddot{q} + \mathbf{C}(q, \dot{q})\dot{q} + \mathbf{G}(q) + \mathbf{F}_c s(\dot{q}) + \mathbf{F}_v \dot{q} = \tau + \mathbf{D}_1, \quad (1)$$

where $\mathbf{F}_c = \text{diag}(F_{c1}, F_{c2}, \dots, F_{cn})$ and $\mathbf{F}_v = \text{diag}(F_{v1}, F_{v2}, \dots, F_{vn})$ are the Coulomb friction coefficient and viscous friction coefficient, respectively. $\mathbf{D}_1 \in \mathbb{R}^n$ is the matrix of modeling uncertainty. The joint control torque $\tau \in \mathbb{R}^n$ is calculated by

$$\begin{aligned} \tau &= \mu F_L, \\ F_L &= P_1 A_1 - P_2 A_2, \end{aligned} \quad (2)$$

where $\mu = \partial l / \partial q$ is the joint Jacobian matrix, $l \in \mathbb{R}^n$ the length of the cylinders, $F_L \in \mathbb{R}^n$ the control thrust of the hydraulic cylinders. $A_1 \in \mathbb{R}^n$ and $P_1 = \text{diag}(p_{11}, p_{12}, \dots, p_{1n})$ represent the area and pressure of the cylinders with rod, $A_2 \in \mathbb{R}^n$ and $P_2 = \text{diag}(p_{21}, p_{22}, \dots, p_{2n})$ the area and pressure of the cylinders without rod. The matrix of inertial $\mathbf{M} \in \mathbb{R}^{n \times n}$, the matrix of Coriolis force and centrifugal force $\mathbf{C} \in \mathbb{R}^{n \times n}$ and the matrix of gravity $\mathbf{G} \in \mathbb{R}^n$ can be written as:

$$\begin{aligned} \mathbf{M}(q) &= \mathbf{M}_1(q) + \mathbf{M}_t(q)m_t, \\ \mathbf{C}(q, \dot{q}) &= \mathbf{C}_1(q, \dot{q}) + \mathbf{C}_t(q, \dot{q})m_t, \\ \mathbf{G}(q) &= \mathbf{G}_1(q) + \mathbf{G}_t(q)m_t, \end{aligned} \quad (3)$$

where $\mathbf{M}_t$, $\mathbf{C}_t$ and $\mathbf{G}_t$ denote additional terms generated due to the unknown load mass. Their detailed expression can be found in the appendix.

According to Ref. [36], the joint dynamics model is expressed as:

$$\begin{aligned} \beta_e^{-1} \dot{F}_L &= -\mathbf{A}_0 \mu \dot{q} + \mathbf{Q}_L + \mathbf{D}_2, \\ \mathbf{A}_0 &= \mathbf{A}_1^2 \mathbf{V}_1^{-1} + \mathbf{A}_2^2 \mathbf{V}_2^{-1}, \end{aligned} \quad (4)$$

where β_e is the elastic bulk modulus of hydraulic oil, and $\mathbf{D}_2 \in \mathbb{R}^n$ is the modeling uncertainty. $\mathbf{Q}_L \in \mathbb{R}^n$ is the valve-controlled flow.

$V_1 = \text{diag}(V_{11}, V_{12}, \dots, V_{1n})$ and $V_2 = \text{diag}(V_{21}, V_{22}, \dots, V_{2n})$ denote the volumes of the cylinders with and without rod respectively.

The following static mapping can describe the relationship between each valve-controlled flow Q_{Li} ($i = 1, 2, \dots, n$) and valve control voltage U_{vi} ($i = 1, 2, \dots, n$) as follows:

$$U_{vi} = \frac{Q_{Li}/k_v}{(A_{1i}/V_{1i})k_{Q1}\sqrt{(\Delta P_{1i})} + (A_{2i}/V_{2i})k_{Q2}\sqrt{(\Delta P_{2i})}}, \quad (i = 1, 2, \dots, n),$$

$$\Delta P_{1i} = \begin{cases} p_{si} - p_{1i}, & U_v \geq 0 \\ p_{1i} - p_{ri}, & U_v < 0 \end{cases}, \quad \Delta P_{2i} = \begin{cases} p_{2i} - p_{ri}, & U_v \geq 0 \\ p_{si} - p_{2i}, & U_v < 0 \end{cases}, \quad (5)$$

where k_{Q1} and k_{Q2} are the flow gain coefficient of the valve, k_v the linearized flow gain of the valve, p_{si} and p_{ri} the supply pressure and the reference pressure of the i th cylinder respectively.

2.2. Parameterization of the dynamic model

Parameters such as the mass, moment of inertia, and coefficient of friction of the joints are unknown. Hydraulic systems also have the characteristics of uncertainty and nonlinearity. To facilitate subsequent controller design, the unknown parameters and nonlinearities are parameterized as:

$$\begin{cases} \theta = [m_t, \beta_e^{-1}, \vartheta_1, \vartheta_2, \vartheta_3, \vartheta_4, \vartheta_5]^T, & \theta \in \mathbb{R}^{5n+2}, \\ \vartheta_i = [\pi_i(i), \dots, \pi_j(i), \dots, \pi_n(i)], & (i = 1, 2, \dots, 5), \\ \pi_j = [m_j, F_{cj}, F_{vj}, D_{1cj}, D_{2cj}], & (j = 1, 2, \dots, n), \end{cases} \quad (6)$$

where $\vartheta_i \in \mathbb{R}^n$ represents the parameter set of the i th physical parameter in n DOF, $\pi_j \in \mathbb{R}^6$ represents the parameter set of the j th DOF and m_j ($j = 1, 2, \dots, n$) is the mass of the j th link. For parameters that only appear once in the parameter set, defining $\theta_1 = m_t$, $\theta_2 = \beta_e^{-1}$. In Eq. (6), $D_{ic} = D_i - \Delta D_i = [D_{ic1}, D_{ic2}, \dots, D_{icn}]$ ($i = 1, 2$) is the low frequency calculable part of D_i .

The general dynamics and the hydraulic dynamics of the manipulator have the following properties.

Property 1

For a certain form of θ , the general dynamics and the hydraulic dynamics of the manipulator can be expressed in the form of linear regression as follows:

$$\tau = \varphi_1^T \theta + \Delta D_1, \quad (7)$$

$$\theta_2 \dot{F}_L = -A_0 \mu \dot{q} + Q_L + \varphi_2^T \theta + \Delta D_2, \quad (8)$$

where φ_i ($i = 1, 2$) are the linear regression matrixes, ΔD_i ($i = 1, 2$), the incalculable part of D_i .

Due to the boundary limitations of the above unknown quantities in practice, the following assumptions can be made:

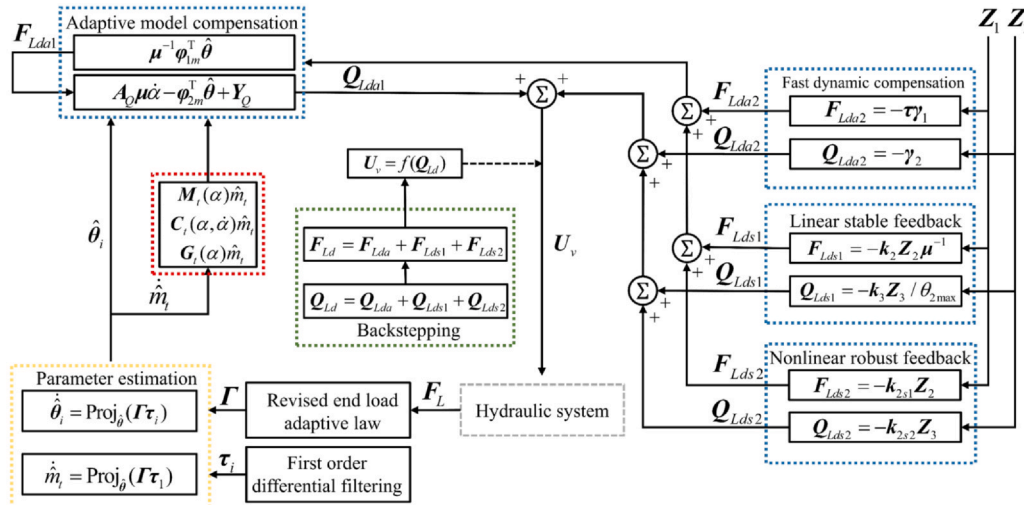


Fig. 2. Direct/Indirect adaptive robust controller (DIARC). The adaptive law is marked with yellow frame and the control law is marked with blue frame. Green frame indicates the usage of backstepping strategy and red frame indicates manipulator dynamics. The controller uses data from hydraulic cylinder pressure sensors and joint angle encoders as inputs.

$$\begin{aligned} \theta_i \in \Omega_{\theta_i} &\triangleq \{\theta_i: \theta_{i\min} \leq \theta_i \leq \theta_{i\max}\}, \quad (i = 1, 2, \dots, 5n + 2), \\ \|\Delta D_i\| &\leq \delta_i, \quad (i = 1, 2, 3), \end{aligned} \quad (9)$$

where δ_i is a known vector. And all parameters are unknown but slowly changing or constant.

2.3. Control objective

The first target is in this paper to generate the valve control voltage U_{vi} in Eq. (5). In the presence of noise and unknown load mass, the tracking accuracy of each joint of the manipulator can still be achieved. The second target is to make precise estimation of the unknown load mass in a comparatively short time, i.e., to make θ_1 converge to its real value.

3. Control of the manipulator

3.1. Controller design

The algorithm diagram of DIARC controller is shown in Fig. 2. Backstepping strategy is adopted to cope with the uncertainty and nonlinearity of hydraulic dynamics.

Step 1

The angular error is defined as:

$$Z_1 = q - q_d \in \mathbb{R}^n, \quad (10)$$

where $q \in \mathbb{R}^n$ is the actual angle of each joint, and $q_d \in \mathbb{R}^n$ the desired angle of each joint.

The angular velocity error is thus defined as:

$$Z_2 = \dot{Z}_1 + k_1 Z_1 = \dot{q} - \dot{q}_{eq}, \quad \dot{q}_{eq} = \dot{q}_d - k_1 Z_1, \quad (11)$$

$$\dot{Z}_2 = \ddot{q} - \ddot{q}_{eq}, \quad (12)$$

where $Z_2 \in \mathbb{R}^n$ and $k_1 \in \mathbb{R}^{n \times n}$ are the positive definite diagonal matrices.

Then, the control object is to make Z_2 converge to zero. Based on Property 1, Eq. (1) can equally be written as:

$$\begin{aligned} M(q) \ddot{q}_{eq} + C(q, \dot{q}) \dot{q}_{eq} + G(q) + F_c s(\dot{q}) + F_s q \\ = \varphi_{1m}^T \theta + D_{1c}, \end{aligned} \quad (13)$$

where φ_{1m} is also a linear regression matrix.

Eq. (1) minus Eq. (13) and consider Eq. (2), we can get:

$$M(q) \dot{Z}_2 + C(q, \dot{q}) Z_2 = -\varphi_{1m}^T \theta + \mu F_L + \Delta D_1. \quad (14)$$

Let $\hat{\cdot}$ represent the estimation of $\cdot$, and $\tilde{\cdot} = \hat{\cdot} - \cdot$ is the error between the real value and the estimation. Similar to tracking error, we further design

virtual control thrust F_{Ld} , which can be dismantled into the following parts:

$$\begin{aligned} F_{Ld} &= F_{Lda} + F_{Lds}, \\ F_{Lda} &= F_{Lda1} + F_{Lda2}, \quad F_{Lds} = F_{Lds1} + F_{Lds2}, \\ F_{Lda1} &= \mu^{-1} \varphi_{1m}^T \hat{\theta}, \quad F_{Lds1} = -k_2 \mu^{-1} Z_2, \end{aligned} \quad (15)$$

where $F_{Ld} \in \mathbb{R}^n$ is the virtual force input. F_{Lda1} is designed to compensate nonlinearities. F_{Lda2} is the fast-dynamic compensation item which will be designed in Step 2. F_{Lds1} can keep the system stable if $k_2 \in \mathbb{R}^{n \times n}$ is a positive definite diagonal matrix. F_{Lds2} is set to compensate the parameter uncertainties, which be designed in Step 2. If we nominate the force tracking error as Eq. (16), according to Eqs. (14) and (15), the derivatives of Z_2 are written as:

$$Z_3 = F_L - F_{Ld}, \quad Z_3 \in \mathbb{R}^n, \quad (16)$$

$$\begin{aligned} M(q) \dot{Z}_2 + C(q, \dot{q}) Z_2 &= -\varphi_{1m}^T \tilde{\theta} + \mu Z_3 - k_2 Z_2 \\ &+ \mu(F_{Lda2} + F_{Lds2}) + \Delta D_1. \end{aligned} \quad (17)$$

Step 2

Based on Eq. (8), the derivatives of Z_3 can be further expressed as:

$$\begin{aligned} \theta_2 \dot{Z}_3 &= \theta_2 \dot{F}_L - \theta_2 \dot{F}_{Ld} \\ &= -A_Q \mu \dot{q} + Q_L + \varphi_2^T \tilde{\theta} - \theta_2 \dot{F}_{Ld} + \Delta D_2, \end{aligned} \quad (18)$$

where $\dot{F}_{Ld}$ can be dismantled into the calculable part $\dot{F}_{Ldc}$ and the in-calculable part $\dot{F}_{Ldi}$, that is:

$$\begin{aligned} \dot{F}_{Ld} &= \dot{F}_{Ldc} + \dot{F}_{Ldi}, \\ \dot{F}_{Ldc} &= \frac{\partial F_{Ld}}{\partial q} \dot{q} + \frac{\partial F_{Ld}}{\partial \dot{q}} \ddot{q} + \frac{\partial F_{Ld}}{\partial t}, \\ \dot{F}_{Ldi} &= \frac{\partial F_{Ld}}{\partial \hat{q}} (\dot{\hat{q}} - \dot{q}) + \frac{\partial F_{Ld}}{\partial \hat{\theta}} \dot{\hat{\theta}}, \end{aligned} \quad (19)$$

where $\hat{q}$ is the estimated value of q .

If we consider Eqs. (18) and (19) can be expressed as:

$$\theta_2 \dot{Z}_3 = -A_Q \mu \dot{q} + Q_L + \varphi_{2m}^T \tilde{\theta} - \theta_2 \dot{F}_{Ldi} + \Delta D_2. \quad (20)$$

Similar to the design of virtual force input, virtual flow input $Q_{Ld} \in \mathbb{R}^n$ is written as:

$$\begin{aligned} Q_{Ld} &= Q_{Lda} + Q_{Lds}, \\ Q_{Lda} &= Q_{Lda1} + Q_{Lda2}, \quad Q_{Lds} = Q_{Lds1} + Q_{Lds2}, \\ Q_{Lda1} &= A_Q \mu \dot{q} - \varphi_{2m}^T \tilde{\theta} + Y_Q, \quad Q_{Lds1} = -\frac{k_3}{\theta_{2\max}} Z_3. \end{aligned} \quad (21)$$

In Eq. (21), Q_{Lda1} is designed to compensate nonlinearities, while Q_{Lda2} is the fast-dynamic compensation item. Q_{Lds1} can keep the system stable if $k_3 \in \mathbb{R}^{n \times n}$ is a positive definite diagonal matrix. Q_{Lds2} is set to compensate the parameter uncertainties, which will be designed later. $Y_Q = -\mu Z_2 \omega_1 / \omega_2$ is the backstepping compensating item. Due to the fact that Z_2 and Z_3 may not be of the same order of magnitude in practice, we introduce $\omega_1 = \text{diag}(\omega_{11}, \dots, \omega_{1n}) \in \mathbb{R}^{n \times n}$ and $\omega_2 = \text{diag}(\omega_{21}, \dots, \omega_{2n}) \in \mathbb{R}^{n \times n}$ to balance their magnitudes and compensate the extra item in subsequent Lyapunov stability analysis. Based on Eqs. (20) and (21), $\dot{Z}_3$ can be described as:

$$\begin{aligned} \theta_2 \dot{Z}_3 &= -\varphi_{2m}^T \tilde{\theta} + k_3 Z_3 \frac{1}{\theta_{2\max}} - Y_Q + Q_{Lda2} \\ &+ Q_{Lds2} - \theta_2 \dot{F}_{Ldi} + \Delta D_2. \end{aligned} \quad (22)$$

Since k_3 is a positive definite diagonal matrix. Z_3 can converge to zero if the small quantities are ignored.

So far, the stability of the system is ensured and the compensation of nonlinearities is completed. Now, design F_{Lda2} and Q_{Lda2} as fast-dynamic compensation items. They will play a significant role in reducing the dynamic error. The modeling uncertainty and parameter estimation error in Eqs. (17) and (22) can be divided into the low frequency part $\tilde{d}_i \in \mathbb{R}^n$ and high frequency part $\tilde{d}_i \in \mathbb{R}^n (i = 1, 2)$, that is:

$$\begin{cases} \tilde{d}_1 + \tilde{d}_1^* = -\varphi_{1m}^T \tilde{\theta} + \Delta D_1, \\ \tilde{d}_2 + \tilde{d}_2^* = -\varphi_{2m}^T \tilde{\theta} - \theta_2 \dot{F}_{Ldi} + \Delta D_2. \end{cases} \quad (23)$$

Fast dynamic compensation items can thus be designed as:

$$\begin{cases} F_{Lda2} = -\mu^{-1} \tilde{d}_1, \\ Q_{Lda2} = -\tilde{d}_2, \end{cases} \quad (24)$$

where $\tilde{d}_i$ is the estimation of d_i . $\tilde{d}_i$ can be updated by the following adaptation law:

$$\dot{\tilde{d}}_i = \text{Proj}(\gamma_i Z_{i+1}) = \begin{cases} 0, & \text{if } |\tilde{d}_i| = d_{iM} \text{ and } \tilde{d}_i^T(t) Z_{i+1} > 0, \\ \gamma_i Z_{i+1}, & \text{else,} \end{cases} \quad (25)$$

where γ_i is a positive definite diagonal matrix and d_{iM} is the upper bound of d_i . Eqs. (17) and (22) can then be written as:

$$\begin{cases} M_2 \dot{Z}_2 + C_2 Z_2 = \mu Z_3 - k_2 Z_2 + \mu F_{Lds2} - \tilde{d}_1 + \tilde{d}_1^*, \\ \theta_2 \dot{Z}_3 = -k_3 Z_3 \frac{1}{\theta_{2\max}} + Q_{Lds2} - Y_Q - \tilde{d}_2 + \tilde{d}_2^*. \end{cases} \quad (26)$$

F_{Lds2} and Q_{Lds2} must satisfy the following conditions:

Condition 1

$$\begin{aligned} Z_2^T \mu F_{Lds2} &\leq 0, \\ Z_3^T Q_{Lds2} &\leq 0. \end{aligned} \quad (27)$$

Condition 2

$$\begin{aligned} Z_2^T (\mu F_{Lds2} - \tilde{d}_1 + \tilde{d}_1^*) &\leq \eta_1, \\ Z_3^T (\mu Q_{Lds2} - \tilde{d}_2 + \tilde{d}_2^*) &\leq \eta_2, \end{aligned} \quad (28)$$

where η_1 and η_2 are constants. One possible form of F_{Lds2} and Q_{Lds2} can therefore be expressed as:

$$\begin{cases} F_{Lds2} = -k_{2s1} Z_2, \quad k_{2s1} = \frac{1}{4\eta_1} (d_{1M} + \|\theta_M\| \|\varphi_{1m}\| + \nu_1)^2, \\ Q_{Lds2} = -k_{2s2} Z_3, \quad k_{2s2} = \frac{1}{4\eta_2} (d_{2M} + \|\theta_M\| \|\varphi_{2m}\| + \nu_2)^2, \end{cases} \quad (29)$$

where θ_M is the upper limit of θ and $\nu_i (i = 1, 2)$ are the positive definite diagonal matrices.

By now the design of Q_{Ld} has finished and the valve control voltage U_{vi} can be obtained from Eq. (5).

3.2. Least squares parameter estimation

In the previous steps, Q_{Ld} is designed to replace Q_L , and it is enough to ensure basic tracking accuracy. In this part, a least squares parameter estimation law is proposed. Due to the method's characteristics of exponential convergence, high accuracy and fast computational speed, it is suitable for estimating m_i .

The virtual control torque $u_{si} (i = 1, 2)$ can be expressed as:

$$u_{si} = -\varphi_{im}^T \theta_{si}, \quad (i = 1, 2), \quad (30)$$

where

$$\begin{cases} \varphi_{1m}^T = \varphi_1^T, \\ \varphi_{2m}^T = [-\dot{F}_L, B_Q, C_Q], \\ \theta_{s1}^T = [\theta_1, \theta_3, \theta_4, \dots, \theta_{3n+2}], \\ \theta_{s2}^T = [\theta_2, \theta_{3n+3}, \theta_{3n+4}, \dots, \theta_{5n+2}], \\ u_{s1} = \mu F_L, \\ u_{s2} = Q_L - \mu A_Q \dot{q}. \end{cases}$$

To eliminate the noise caused by insufficient sensor accuracy, a first-order differential device is designed here. It has the following property.

Property 2

The first-order differential device has the following form:

$$\begin{aligned} \sigma \dot{x}_f + x_f &= x, \\ x_{f\text{next}} &= \dot{x}_f t_s + x_f, \end{aligned} \quad (31)$$

where x_f is the filtered x , $x_{f\text{next}}$ is the updated x_f , σ is the time constant selected for first-order differential filtering and t_s is the sampling time. It is

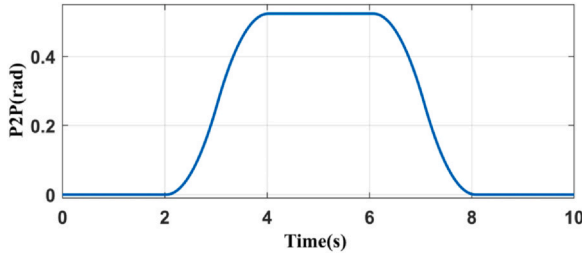


Fig. 3. P2P trajectory used in the experiment and simulation.

noteworthy that when the initial condition, denoted as $x(0) \neq 0$, an exponential decay term, represented by $e^{-t/\sigma}x(0) \rightarrow 0$ with $t \rightarrow \infty$, emerges. The presence of this diminishing term does not substantially influence the principal outcomes of the convergence analysis. Consequently, for the sake of simplicity, the influence of this initial condition will be omitted in the subsequent analysis. After filtering, the first equation in Eq. (30) is described as Eq. (32) and its estimation error is expressed as Eq. (33).

$$\mathbf{u}_{sif} = -\tilde{\varphi}_{if}^T \hat{\theta}_{si}, \quad (i = 1, 2), \quad (32)$$

$$\varepsilon_i = \hat{\mathbf{u}}_{sif} - \mathbf{u}_{sif} = -\tilde{\varphi}_{if}^T \tilde{\theta}_{si}, \quad (i = 1, 2). \quad (33)$$

The next step is to design a least square adaptive law which can enable ε_i to converge to zero exponentially within a finite time. The positive definite gain matrix Γ_i ($i = 1, 2$), can be expressed as:

$$\dot{\Gamma}_i = \begin{cases} \alpha_i \Gamma_i - \zeta_i \tilde{\varphi}_{if} \tilde{\varphi}_{if}^T \Gamma_i, & \text{if } \lambda_i \max(\Gamma_i(t)) \leq \rho_{im}, (i = 1, 2), \\ 0, & \text{otherwise,} \end{cases} \quad (34)$$

where α_i is the forgetting factor and ρ_{im} is the upper bound of the projective adaptive law, $\zeta_i = (1 + \mathbf{v}_i^T \tilde{\varphi}_{if} \Gamma_i \tilde{\varphi}_{if} \mathbf{v}_i)^{-1}$, and all the elements in $\mathbf{v}_i \in \mathbb{R}^n$ are positive. Then, the projective adaptive law is written by:

$$\dot{\hat{\theta}}_{si} = \text{Proj}_{\hat{\theta}}(\Gamma \tau), \quad \tau = \zeta_i \tilde{\varphi}_{if} \varepsilon_i. \quad (35)$$

The projection function that satisfies Eq. (9) should be designed to bound the uncertain parameters. One of the possible choices would be:

$$\text{Proj}_{\hat{\theta}}(\cdot) = \begin{cases} \cdot, & \text{if } \hat{\theta} \in \partial \hat{\Omega}_{\theta} \text{ or } (\mathbf{n}_{\hat{\theta}}^T \cdot) \leq 0, \\ \left(\mathbf{I} - \Gamma \frac{\mathbf{n}_{\hat{\theta}} \mathbf{n}_{\hat{\theta}}^T}{\mathbf{n}_{\hat{\theta}}^T \Gamma \mathbf{n}_{\hat{\theta}}} \right) \cdot, & \text{if } \hat{\theta} \in \partial \Omega_{\theta} \text{ and } (\mathbf{n}_{\hat{\theta}}^T \cdot) > 0, \end{cases} \quad (36)$$

where $\hat{\Omega}_{\theta}$ and Ω_{θ} denote the interior and boundaries of the parameter sets, respectively. $\mathbf{n}_{\hat{\theta}}$ is the unit normal vector pointing to the external world when $\hat{\theta}$ is at the boundary.

Eq. (36) is used to update m_i . To achieve fast parameter adaptation effect, parameters should be carefully adjusted. It is noteworthy that α_1 should be as small as possible to minimize the impact of older data on new data. Conversely, Γ_1 should be large enough so that the speed of parameter convergence can be maximized.

Due to the frequent changes in unknown load mass under actual working conditions, if m_i cannot be updated quickly, the control accuracy will be difficult to ensure. Eq. (36) indicates how Γ_1 is updated. However, it cannot guarantee that Γ_1 can be large enough to quickly update m_i . When the load of the manipulator changes, it can be considered that the adaptive value of m_i has fallen into local optima. At this time, Γ_1 should be modified to drive the adaptation of m_i and make it jump out of local optima.

The objective function is shown as:

$$\Delta F = F_L - F_{nLda1}, \quad (37)$$

where $F_L = p_{1n} A_{1n} - p_{2n} A_{2n}$ and F_{nLda1} is the last item in F_{Lda1} . Subsequently, Γ_1 follows the modification law expressed as:

$$\Gamma_1 = \begin{cases} \bar{\Gamma}_1, & \Delta F > F_{\max} \text{ or } \Delta F < F_{\min}, \\ \Gamma_1, & \text{else,} \end{cases} \quad (38)$$

where $\bar{\Gamma}_1$ is the original parameter of Γ_1 , $F_{\max}$ and $F_{\min}$ are known functions.

Theorem 1

All the signals in the system are bounded. If a positive-definite function is defined as $V_s = \omega_1 \mathbf{Z}_2^T \mathbf{M}_2 \mathbf{Z}_2 / 2 + \omega_2 \theta_2 \mathbf{Z}_3^T \mathbf{Z}_3 / 2$, then the following inequality is satisfied:

$$V_s(t) \leq \exp(-\lambda t) V_s(0) + \frac{\eta}{\lambda} [1 - \exp(-\lambda t)], \quad (39)$$

where $\lambda = 2 \times \min \{ \lambda_{\min}(k_2)/k_M, \lambda_{\min}(k_3)/\theta_{2\max}^2 \}$, $k_M \mathbf{I}_n \geq \mathbf{M}(q)$.

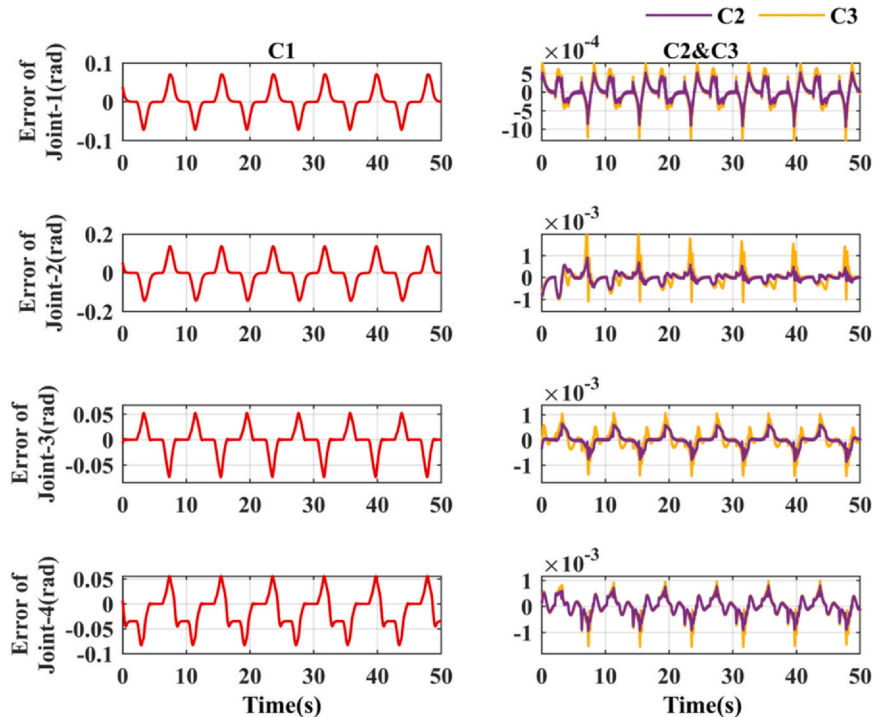


Fig. 4. Simulation results on tracking accuracy.

Table 1
Coefficients of DIRAC with load compensation.

Coefficients	Value
k_1	$[100, 70, 55, 90]^T$
k_2	$[35, 45, 70, 75]^T$
k_3	$[40, 80, 85]^T$
γ_1	$[220, 280, 150, 120]^T$
γ_2	$[6 \times 10^{-7}, 8.6 \times 10^{-7}, 9 \times 10^{-8}, 3 \times 10^{-7}]^T$

Theorem 2

If the persistent excitation condition is satisfied, $\hat{\theta}$ is able to converge to its real value in the presence of parameter uncertainty only. Based on Theorem 1, zero angular tracking error can be achieved, i.e., $Z_1 \rightarrow 0$ when $t \rightarrow \infty$. And the proof of Theorem 1 and Theorem 2 can be found in Appendix.

Theorem 1 indicates that the controller possesses exponential converging rate. Theorem 2 demonstrates that uncertain parameters such as m_i can be estimated online and the negative impact of parameter uncertainty can be eliminated through parameter estimation.

4. Simulation

In the simulation, a four-DOF hydraulic manipulator is modeled using MATLAB and Simscape Multibody. The angle coupling, friction and gravity are all taken into consideration. The parameters of the links and hydraulic systems are strictly consistent with the real manipulator. In the simulation, a spherical object with a mass of 10 kg and a radius of 0.04 m is fixed in the gripper of the manipulator. The distance between the center of the weight and the rotation axis of joint-4 is set to 0.4 m. Point-to-point (P2P) trajectory is used in this experiment, as shown in Fig. 3.

The control methods used in the simulation are explained as follows:

C1: PID controller with coefficients being tuned to enhance the tracking accuracy.

C2: DIARC controller with load compensation.

C3: DIARC controller without load compensation. Similar to C2, but the unknown load m_i will not be compensated, i.e., $\theta_i = 0$. The other parameters are the same as C2.

Table 2
Control performance of each controller.

Controller	Angle	Maximum error ($\times 10^{-4}$ °)	RMSE ($\times 10^{-6}$ °)	IAE (°·s)
C1	Joint-1	636	300	1210
	Joint-2	1819	5100	5378
	Joint-3	2067	6800	6350
	Joint-4	1135	1200	2687
C2	Joint-1	77	4.304	201.5
	Joint-2	107	8.358	261.98
	Joint-3	120	7.538	202.56
	Joint-4	99	5.01	194.28
C3	Joint-1	115	8.4	264.89
	Joint-2	151	8.367	279.13
	Joint-3	139	10.97	261.21
	Joint-4	155	18.43	352.47

Fig. 4 shows the tracking error of the three methods. The control accuracy using C1 is the lowest, with the maximum error of joint-2 reaching 0.13 rad. C3 can achieve much better performance than PID controller. C2 has the best control accuracy in all four joints, that the max errors decrease by 29.75%, 67.25%, 43.96% and 40.04%, respectively.

5. Experiment setup

5.1. Hardware platforms

The tested four-DOF manipulator is shown in Fig. 1(c). The hydraulic system of the manipulator has the pump driving motor (ABB M2BAX-112MA-4), with rated power being 4 kW and maximum output flowrate being 12 L/min. It installs two pressure sensors (GEFRAN-KS-E-Z-B16D M-V-530) to measure the pressure of each chamber. There are four angle sensors (Melexis-MLX90316) located on each joint. Quarc QPID-e and Q8-USB are utilized to process the data collected by the sensors. The overall sampling time is set to be 25 ms.

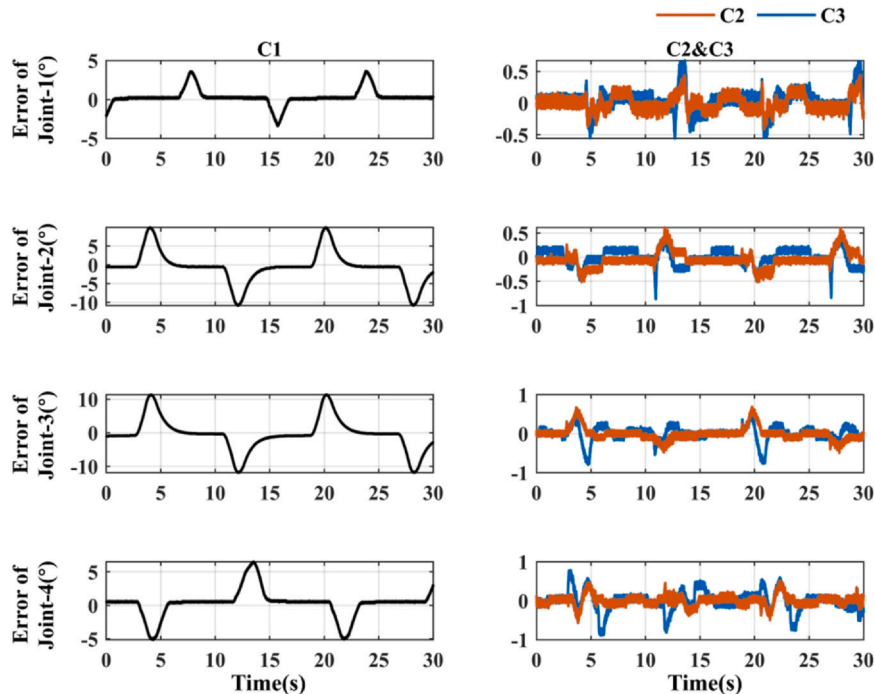


Fig. 5. Experiment results on tracking accuracy.

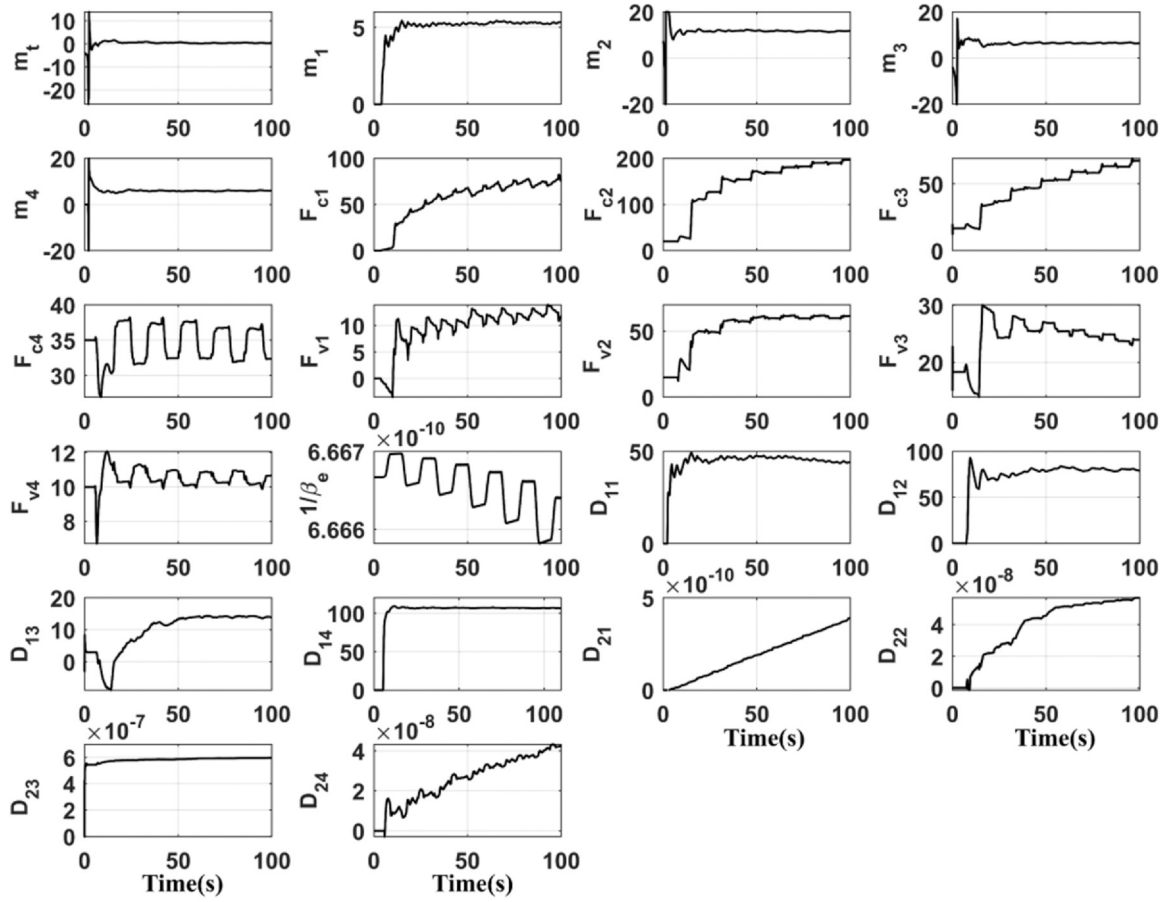


Fig. 6. Adaptive curves of all parameters.

5.2. Experiment design

To verify the effectiveness of the algorithm for precise motion control and precise load estimation, two sets of experiments are designed as follows.

Set 1: PID trajectory tracking experiment. The end of the manipulator grabs a heavy object with unknown mass, and each joint tracks the P2P trajectory while changing the load mass multiple times to test the effectiveness of the proposed algorithm for precision control and load estimation.

Set 2: Simulation of common handling tasks. The manipulator grabs a heavy object at starting point A and transports it to the designated end B, while estimating the mass of the unknown load in real-time.

Using the parameters in simulation as reference, the coefficients in the experiment are set as follows.

C1: PID controller. The control coefficients are $k_p = [6, 10, 14, 44]$, $k_i = [2, 2.5, 1.5, 4]$, $k_d = [0.5, 0.3, 0.5, 1]$.

C2: DIARC controller with load compensation. Key parameters related to control accuracy and load estimation are shown in Table 1. They have been adjusted during the experiment process to improve control and estimation effects. All initial values of parameter estimation are set to zeros

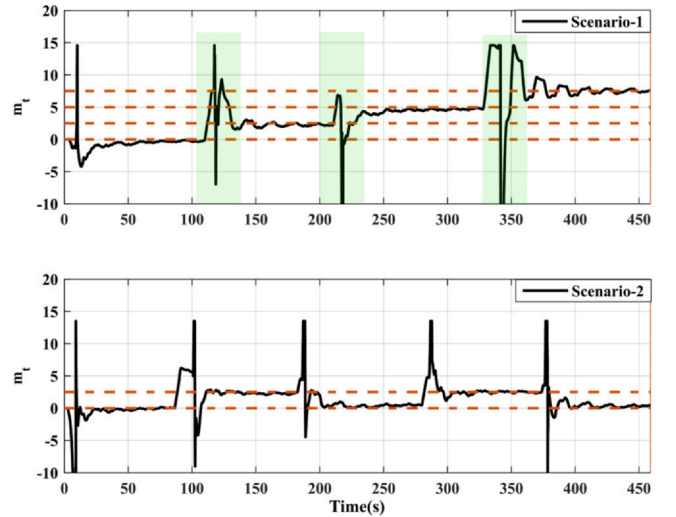


Fig. 7. Adaptive curves of unknown load estimation.

Table 3
Statistic data in load estimation accuracy.

Mass (kg)	Avg. (kg)	Max. (kg)	Min. (kg)	RMSE (kg)	SD (kg)	Error (%) (Avg./Max./Min.)
0	0.175	0.35	-0.1	0.018	0.142	
2.5	2.607	2.74	2.42	0.015	0.127	4.3/9.6/3.2
5	5.14	5.33	4.80	0.062	0.263	2.8/6.6/4.0
7.5	7.71	8.12	7.33	0.055	0.247	2.8/8.3/2.3

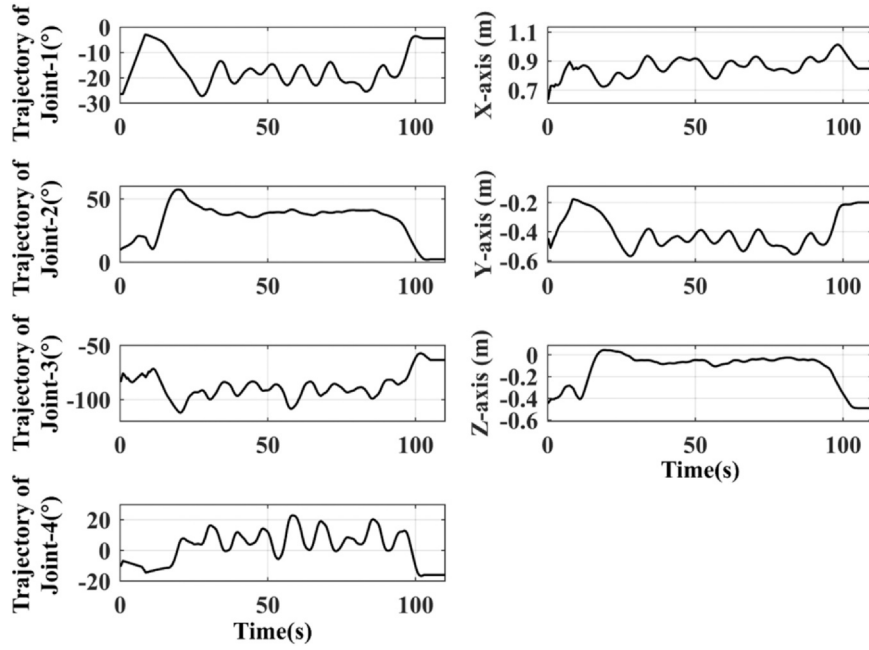


Fig. 8. Desired trajectory for experiment Set-2.

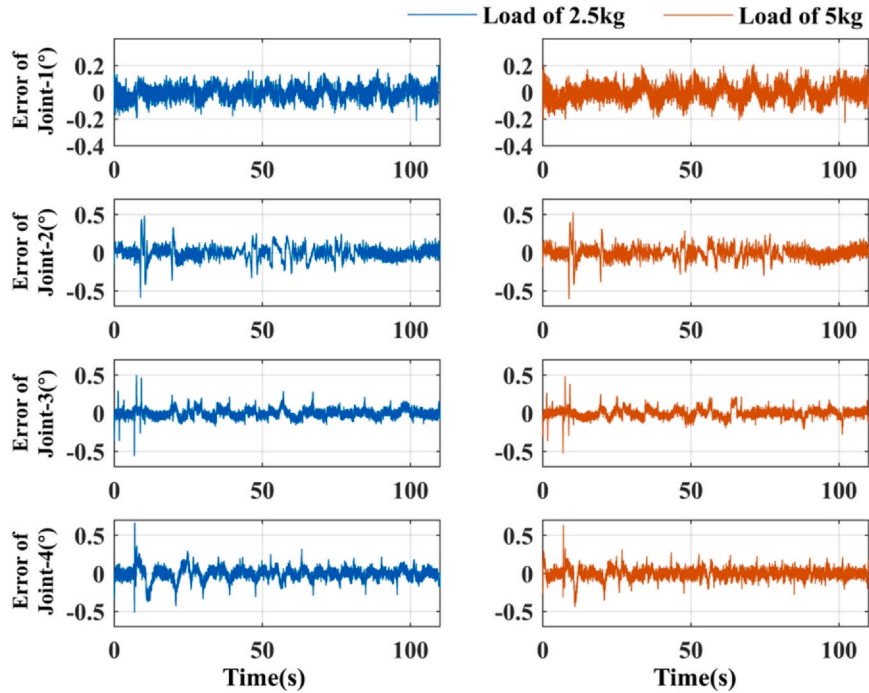


Fig. 9. Tracking performance in joint space.

C3: DIARC controller without load compensation. $\theta_1 = 0$ while other coefficients and the initial values of parameter estimation are the same as C2.

6. Experiment results

Set 1

Fig. 5 shows the tracking performance of each controller. Table 2 lists the maximum tracking error, root mean squared error (RMSE) and integral absolute error (IAE). It can be seen that C1 is far inferior in control accuracy to C2 and C3 because the complex dynamics of the hydraulic manipulator is considered. For DIARC controllers, load compensation also has a significant effect. In C2, the control accuracy of

joint-2, joint-3 and joint-4 which is directly affected by the unknown load mass has greatly improved. Compared to C3, the maximum errors of the joints decrease by 29.14%, 13.67% and 36.13%, respectively. Due to the high estimation accuracy of the unknown load mass, the maximum error of the joint-1 also reduces by 33.04%. RMSE and IAE of all joints also decrease evidently.

For C2, accurate parameter estimation is an important feature. The adaptive curves of all parameters of the controller are shown in Fig. 6. It demonstrates that each parameter converges to a certain value and the convergence value is largely trustworthy.

Additionally, the manipulator is operated under no-load, loads of 2.5 kg, 5 kg, and 7.5 kg respectively. For each unknown load mass (including no-load), the manipulator runs repeatedly and collect ten

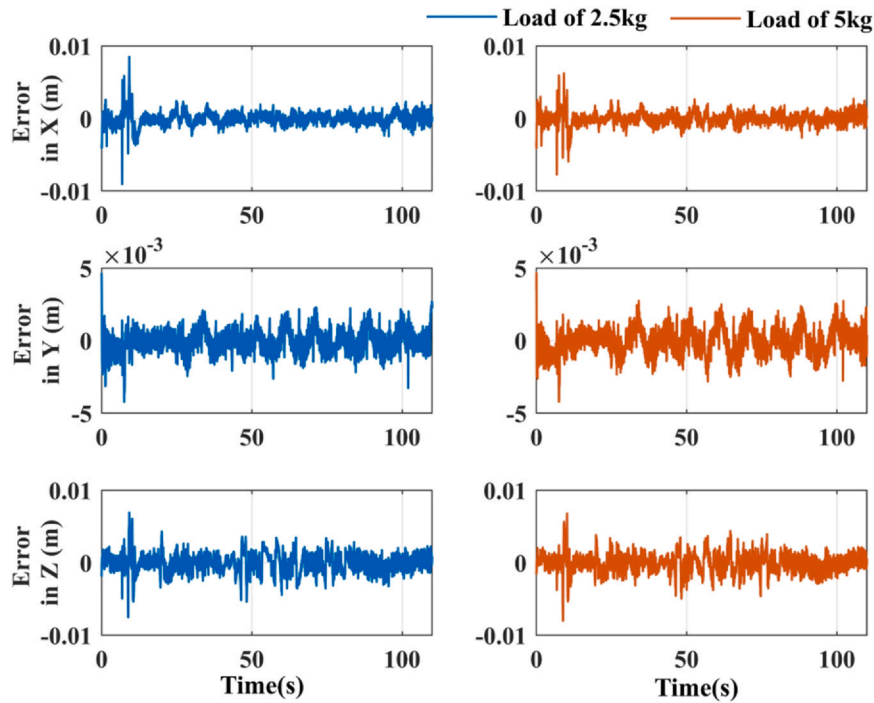


Fig. 10. Tracking performance in Cartesian space.

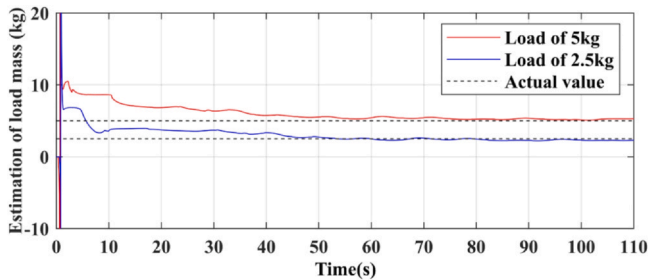


Fig. 11. Adaptive curves in handling task.

sets of data. The average, maximum, and minimum values, RMSE and standard deviation (SD) list in Table 3.

From the experimental data, the average estimation error is always below 5%, and the maximum error does not exceed 10%. Although the estimated values of mass witness certain fluctuation, with the maximum difference reaching 0.79 kg, the average estimated value is close to the actual value. It depicts that the developed controller has an ability to achieve estimation accuracy.

In practical engineering, the unknown load mass of a manipulator is often time-varying. Two scenarios are set up to test the controller's real-time estimation of changing loads. In scenario-1, the unknown load mass continues to increase by 0 kg, 2.5 kg, 5 kg, and 7.5 kg, respectively. In scenario-2, the unknown load mass switches between 0 kg and 2.5 kg. The adaptive curves are shown in Fig. 7.

Evidently, in both scenarios, the adaptive curve converges to the actual values. In the adaptive curve of scenario-1, the intermediate transition state is marked with backup color. During this period, the load is changed and Γ_1 modification (expressed in Eq. (38)) is performed, which is indicated by a spike of the adaptive curve. The duration of this transition state is short, and the parameters will continue to converge to the true value afterwards. The time interval for estimation is short with 10–15 s on average and the average estimation error in the experiment is 3.89%.

Set 2

In this experiment, a common handling task is performed. The manipulator carries a weight of 2.5 kg and 5 kg respectively. The trajectory in joint

space and Cartesian space is shown in Fig. 8. The tracking results in joint space are shown in Fig. 9. The maximum tracking error occurs at around 9 s when the angular velocity witnesses a significant change. But this transient error does not exceed 0.7° , and in most cases, the error is limited to within 0.2° . Additionally, the tracking results in three dimensions of the Cartesian space are manifested in Fig. 10. Generally, the error in all directions can be stabilized within 0.5 cm, and the maximum error is limited to within 1 cm. Meanwhile, it is noted that when the load mass changes from 2.5 kg to 5 kg, there is no significant change in maximum error and dynamic error, indicating that the compensation for the load mass is effective. To conclude, the designed controller can compensate for unknown load mass and achieve high control accuracy.

The adaptive curves for the two handling tasks are illustrated in Fig. 11. The system feels the heavy object being picked up at the beginning and activates the modification of Γ_1 , which results in the spike of the curve. At 10 s, the curve becomes relatively stable and gradually converges to the actual values, and it takes around 40 s to truly converge to the actual value. In this experiment, the frequency width of velocity and acceleration is relatively limited, so it is not possible to fully stimulate the dynamic characteristics of the system. Due to the strong correlation between load mass and system velocity and acceleration, the estimation speed of mass is slowed down, but high estimation accuracy is still achieved. A mass of 2.35 kg is estimated when the actual weight is 2.5 kg, and a mass of 5.22 kg is estimated when the actual weight is 5 kg. The estimation error is 6% and 4.4%, respectively.

7. Conclusions

- (1) A direct/indirect adaptive robust controller (DIARC) is developed, which can compensate for the influence of the unknown payload on hydraulic manipulators and can effectively estimate the payload mass. The high-order dynamics of the hydraulic manipulator with complex nonlinearity can be solved by adopting backstepping strategy which enables the controller to achieve high tracking accuracy.
- (2) A dynamic model is proposed with the unknown load mass being included in the parametric space. A parameter adaptive law based on least squares method is designed in response to the demand for online identification of unknown load mass, and the precise estimation of

unknown load mass can be ensured. With the accurate estimation of the unknown load mass, the control accuracy can be further improved.

- (3) The control performance is experimentally verified on a four-DOF hydraulic manipulator. The experimental results show that the results demonstrate that, under a 5 kg load, the average control precision across all four joints improved by 28.0% compared to controllers without load compensation. Repeated experiments reveal that the average precision of load estimation is below 4.3%, with accuracy improving to less than 2.8% under heavier load conditions. Overall, there is a trend indicating that larger loads result in higher estimation accuracy. Future works include adopting new adaptive laws to further enhance the estimation accuracy and speed, as well as validating online estimation capabilities under full working conditions.

Authors' contributions

Qixian Wang proposed the methodology and conducted experiments, he also drafted the original manuscript. Manzhi Qi helped with the experiments and utilized software for simulation, he also did

investigation on the utilized method. Yangxiu Xia helped to organize experiments and conducted data curation. Shizhao Zhou supervised the experiments and helped to polish the paper. Zheng Chen supervised the entire research process and offered resources, he is also the administrator of the project and related funding. All authors read and approved the final manuscript.

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Declaration of Competing Interest

The authors declare no competing financial interests.

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Not applicable

Appendix A

Detailed expression of M_i , C_i and G_i are expressed as:

$$\begin{aligned} M_i &= \begin{pmatrix} M_{11} & 0 & 0 & 0 \\ 0 & M_{22} & M_{23} & M_{24} \\ 0 & M_{32} & M_{33} & M_{34} \\ 0 & M_{42} & M_{43} & M_{44} \end{pmatrix}, \quad C_i = \begin{pmatrix} C_{11} & C_{12} & C_{13} & C_{14} \\ C_{21} & C_{22} & C_{23} & C_{24} \\ C_{31} & C_{32} & C_{33} & C_{34} \\ C_{41} & C_{42} & C_{43} & 0 \end{pmatrix}, \\ G_i &= \begin{pmatrix} 0 \\ G_2 \\ G_3 \\ G_4 \end{pmatrix}, \end{aligned} \quad (40)$$

where

$$\begin{aligned} M_{11} &= (L_1 + L_3 C_{23} + L_2 C_2 + L_4 C_{234})^2, \\ M_{22} &= M_{33} + L_2^2 + 2(C_2 L_2 L_3 + C_{34} L_2 L_4), \\ M_{23} &= M_{32} = M_{33} + (2C_{23} L_2 L_4 + C_3 L_2 L_3), \\ M_{24} &= M_{42} = M_{34} + L_4 L_2 C_{34}, \\ M_{33} &= M_7 + (L_3^2 + 2C_4 L_3 L_4), \\ M_{34} &= M_{43} = L_4 (L_3 C_4 + L_4), \\ M_{44} &= L_4^2, \quad x = (L_1 + L_2 C_2 + L_3 C_{23} + L_4 C_{234}), \\ C_{11} &= -(x - L_1)(\ddot{\alpha}_2 + \ddot{\alpha}_3 + \ddot{\alpha}_4)x, \\ C_{12} &= -L_2 \ddot{\alpha}_1 S_2 x + C_{13}, \\ C_{13} &= -L_3 \ddot{\alpha}_1 S_{23} x + C_{14}, \\ C_{14} &= -L_4 \ddot{\alpha}_1 S_{234} x, \\ C_{21} &= L_2 S_2 \ddot{\alpha}_1 x + C_{31}, \\ C_{22} &= -\ddot{\alpha}_3 (L_2 L_4 S_{34} + L_2 L_3 S_3) - \ddot{\alpha}_4 (L_2 L_4 S_{34} + L_3 L_4 S_4), \\ C_{23} &= -(\ddot{\alpha}_2 + \ddot{\alpha}_3 + \ddot{\alpha}_4) L_2 L_4 S_{34} - (\ddot{\alpha}_2 + \ddot{\alpha}_3) L_2 L_3 S_3 + C_{33}, \\ C_{24} &= C_{34} - L_2 L_4 S_{34} (\ddot{\alpha}_2 + \ddot{\alpha}_3 + \ddot{\alpha}_4), \\ C_{31} &= L_3 S_{23} \ddot{\alpha}_1 x + C_{41}, \\ C_{32} &= L_2 L_4 S_{34} \ddot{\alpha}_2 + L_2 L_3 S_3 \ddot{\alpha}_2 + C_{33}, \\ C_{33} &= -L_3 L_4 S_4 \ddot{\alpha}_4, \\ C_{34} &= C_{33} - C_{43}, \\ C_{41} &= L_4 S_{234} \ddot{\alpha}_1 x, \\ C_{42} &= L_2 L_4 S_{34} \ddot{\alpha}_2 + C_{43}, \\ C_{43} &= L_3 L_4 S_4 (\ddot{\alpha}_2 + \ddot{\alpha}_3), \\ G_2 &= g L_2 C_2 + G_3, \\ G_3 &= g L_3 C_{23} + G_4, \\ G_4 &= g L_4 C_{234}. \end{aligned}$$

Appendix B

Proof for Theorem 1

The Lyapunov function is defined as:

$$V_s = \frac{1}{2}\omega_1 \mathbf{Z}_2^T \mathbf{M}_2 \mathbf{Z}_2 + \frac{1}{2}\omega_2 \theta_2 \mathbf{Z}_3^T \mathbf{Z}_3. \quad (41)$$

Since $\dot{\mathbf{M}}_2 = \mathbf{C}_2 + \mathbf{C}_2^T$, the derivative of Eq. (41) is written as:

$$\begin{aligned} \dot{V}_s &= \omega_1 \mathbf{Z}_2^T (\mathbf{M}_2 \dot{\mathbf{Z}}_2 + \mathbf{C}_2 \mathbf{Z}_2) + \omega_2 \theta_2 \mathbf{Z}_3^T \dot{\mathbf{Z}}_3 \\ &= \omega_1 \mathbf{Z}_2^T (-\varphi_{1m}^T \theta + \mu \mathbf{F}_L + \Delta \mathbf{D}_1) \\ &+ \omega_2 \mathbf{Z}_3^T (-\mathbf{A}_Q \mu \dot{\alpha} + \mathbf{Q}_L + \varphi_{2m}^T \theta + \Delta \mathbf{D}_2). \end{aligned} \quad (42)$$

Considering Eqs. (15) and (21), Eq. (42) is expressed as:

$$\begin{aligned} \dot{V}_s &= -\omega_1 \mathbf{Z}_2^T \mathbf{k}_2 \mathbf{Z}_2 + \omega_1 \mathbf{Z}_2^T (\mu \mathbf{Z}_3 - \tilde{\mathbf{d}}_1 + \tilde{\mathbf{d}}_1^* + \mu \mathbf{F}_{Lds2}) \\ &- \omega_2 \frac{k_3}{\theta_{2\max}} \mathbf{Z}_3^T + \omega_2 \mathbf{Z}_3^T (\mathbf{Y}_Q - \tilde{\mathbf{d}}_2 + \tilde{\mathbf{d}}_2^* + \mathbf{Q}_{Lds2}). \end{aligned} \quad (43)$$

Noting Eq. (29) and $\mathbf{Y}_Q = -\mu \mathbf{Z}_2 \omega_1 / \omega_2$, Eq. (43) is expressed as:

$$\begin{aligned} \dot{V}_s &\leq -\omega_1 \mathbf{Z}_2^T \mathbf{k}_2 \mathbf{Z}_2 - \omega_2 \frac{k_3}{\theta_{2\max}} \mathbf{Z}_3^T + \omega_1 \eta_1 + \omega_2 \eta_2 \\ &\leq -\lambda V_s + \eta. \end{aligned} \quad (44)$$

Then apply the comparison theorem, we can get:

$$V_s(t) \leq \exp(-\lambda t) V_s(0) + \frac{\eta}{\lambda} [1 - \exp(-\lambda t)]. \quad (45)$$

The boundedness of the closed-loop system signal is guaranteed. The relation between the Laplace transform of $\mathbf{Z}_1$ and $\mathbf{Z}_2$ can be written as:

$$\frac{\mathbf{Z}_1(s)}{\mathbf{Z}_2(s)} = \frac{1}{1 + \mathbf{k}_1 s}. \quad (46)$$

Therefore, if $\mathbf{Z}_2$ is bounded, $\mathbf{Z}_1$ will be bounded as well.

Appendix C

Proof for Theorem 2

According to Ref. [36], when $\Delta \mathbf{D}_1 = \Delta \mathbf{D}_2 = 0$ and PE condition is fulfilled, $\tilde{\theta} \rightarrow 0$ when $t \rightarrow \infty$. The Lyapunov function is defined as:

$$V_a = V_s + \frac{1}{2}\omega_1 \hat{\mathbf{d}}_1^T \Gamma_1 \hat{\mathbf{d}}_1 + \frac{1}{2}\omega_2 \hat{\mathbf{d}}_2^T \Gamma_2 \hat{\mathbf{d}}_2. \quad (47)$$

The derivative of Eq. (47) is written as:

$$\begin{aligned} \dot{V}_a &= \omega_1 \mathbf{Z}_2^T (\mathbf{M}_2 \dot{\mathbf{Z}}_2 + \mathbf{C}_2 \mathbf{Z}_2) + \omega_2 \theta_2 \mathbf{Z}_3^T \dot{\mathbf{Z}}_3 + \omega_1 \hat{\mathbf{d}}_1^T \Gamma_1 \dot{\hat{\mathbf{d}}}_1 \\ &\quad + \omega_2 \hat{\mathbf{d}}_2^T \Gamma_2 \dot{\hat{\mathbf{d}}}_2 \\ &= \omega_1 \mathbf{Z}_2^T (\mathbf{M}_2 \dot{\mathbf{Z}}_2 + \mathbf{C}_2 \mathbf{Z}_2) + \omega_2 \theta_2 \mathbf{Z}_3^T \dot{\mathbf{Z}}_3 \\ &+ \omega_1 \hat{\mathbf{d}}_1^T \Gamma_1 (\text{Proj}(\gamma_1 \mathbf{Z}_2) - \mathbf{Z}_2) + \omega_2 \hat{\mathbf{d}}_2^T \Gamma_2 (\text{Proj}(\gamma_1 \mathbf{Z}_3) - \mathbf{Z}_3) \\ &\leq -\omega_1 \mathbf{Z}_2^T \mathbf{k}_2 \mathbf{Z}_2 - \omega_2 \frac{k_3}{\theta_{2\max}} \mathbf{Z}_3^T + \omega_1 \mathbf{Z}_2 \varphi_{1m}^T \tilde{\theta} + \omega_2 \mathbf{Z}_3 \varphi_{2m}^T \tilde{\theta}. \end{aligned} \quad (48)$$

Since $\tilde{\theta} \in L_2[0, \infty)$ and both φ_{1m}^T and φ_{2m}^T are bounded, $\varphi_{1m}^T \tilde{\theta} \in L_2[0, \infty)$ and $\varphi_{2m}^T \tilde{\theta} \in L_2[0, \infty)$. From Eq. (47), we can get $\mathbf{Z}_2 \in L_2[0, \infty)$ and $\mathbf{Z}_3 \in L_2[0, \infty)$. According to barbalat's lemma $\mathbf{Z}_1$ converges to zero when $t \rightarrow \infty$.

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